Lesson 2.4 Guided Notes

Session 1 · Building a Box Plot

I can find the five-number summary of a data set and use it to build a box plot. · 6.SP.B.4

Word / Spanish	Meaning — complete the blank	Example
five-number summary / resumen de cinco números	Min, Q1, _____, Q3 and max.	5, 9, 14, 20, 23
quartile / cuartil	Q1 is the median of the _____ half; Q3 is the median of the upper half.	lower half 5, 9, 11: Q1 = 9
box plot / diagrama de caja	A graph of the five-number summary: a box from Q1 to Q3 and two _____.	box 9 to 20, line at 14

Key idea

With an odd number of values, leave the _____ out of both halves.

Each of the four sections of a box plot holds about one _____ of the data.

Worked example

Times (min): 14, 5, 20, 9, 23, 11, 16. Find the five-number summary.

Order: 5, 9, 11, 14, 16, 20, 23. Median = _____.

Lower half 5, 9, 11: Q1 = _____. Upper half 16, 20, 23: Q3 = _____.

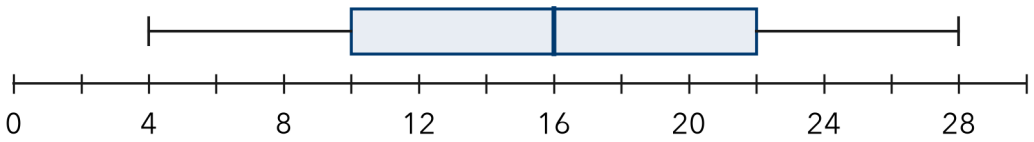
Answer: 5, 9, 14, 20, _____

You try

1. Data: 3, 8, 5, 12, 10, 7, 15. Find each value.
a. Q1: _____
b. Median: _____
c. Q3: _____

_____

2. Read the box plot.



_____

Sum it up: To build a box plot, I first ___, then ___.

_____

Session 2 · Comparing Box Plots

I can compare two groups by reading the medians and box widths of their box plots. · 6.SP.B.4

Word / Spanish	Meaning — complete the blank	Example
center / centro	A typical value; on a box plot, the _____ line.	median 20 min
box width / ancho de la caja	Q3 _____ Q1; the spread of the middle half.	22 − 18 = 4
consistent / constante	Values are _____ together; the box is narrow.	box 18 to 22

Key idea

Compare _____ to see which group has the higher typical value.

Compare box widths: the _____ box is more spread out.

Worked example

Trail X: Q1 18, median 20, Q3 22. Trail Y: Q1 19, median 26, Q3 33 (minutes).

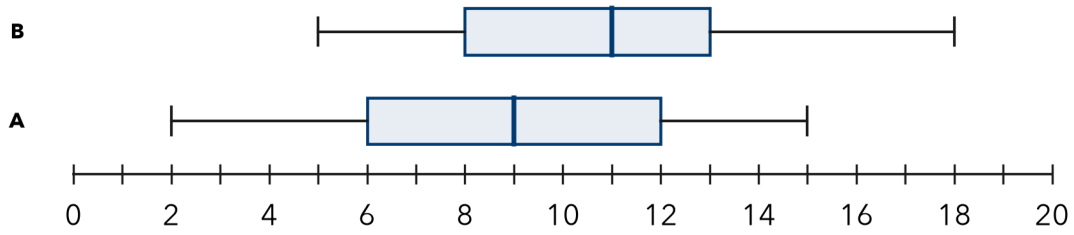
Medians: 26 − 20 = _____, so Y is higher by 6 min.

Box widths: X is _____, Y is _____.

Answer: Trail _____ is more consistent.

You try

1. Which plot has the higher median? Which is more spread out?



_____

2. Find each box width.
a. Q1 = 12, Q3 = 30: _____
b. Q1 = 41, Q3 = 47: _____

_____

Sum it up: To compare two box plots, I look at ___ and ___.

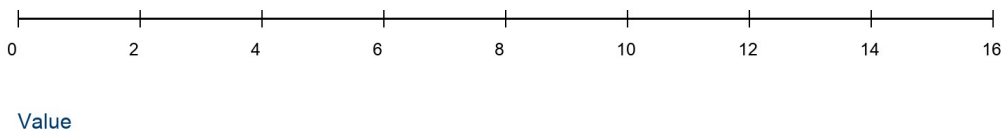
_____

Session 1 · Model, reason, and explain

Building a Box Plot · 6.SP.B.4

Success: represent the quantities, justify the relationship, and check the result with the correct units.

For 2, 4, 6, 8, 10, 12, 14, find the five-number summary and draw a box plot. When there is an odd number of observations, exclude the overall median from both halves.



_____

_____

_____

Extend your reasoning

Explain why the box does not reveal the exact individual observations inside it.

_____

_____

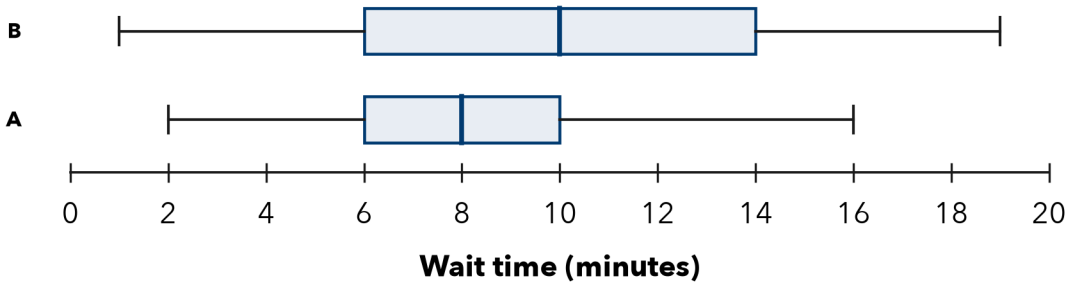
Self-check: Did my model, calculation, explanation, and units agree?

Session 2 · Model, reason, and explain

Comparing Box Plots · 6.SP.B.4

Success: represent the quantities, justify the relationship, and check the result with the correct units.

Compare the two box plots on the same scale. Which has the lower median? Which has the smaller middle-half spread? Can you tell which group has more observations?



_____

_____

_____

Extend your reasoning

Explain why equal-length boxes would not prove that two data sets are identical.

_____

_____

Self-check: Did my model, calculation, explanation, and units agree?